

Mir Rahman

Senior Software Engineer | Cloud & Solutions Architecture

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Professional Summary

AWS-certified Solution Architect and senior software engineer with 12+ years of experience designing and delivering secure, scalable cloud-native systems across financial services, payments, trading, banking, insurance, industrial automation, and IoT. Combines hands-on software engineering with practical solution architecture experience spanning distributed systems, APIs, event-driven integrations, data platforms, security, reliability, and production operations. Skilled at turning business and technical requirements into clear, deployable designs; collaborating with cross-functional stakeholders in different countries; and communicating trade-offs through technical design discussions and documentation.

Technical Skills

- **Programming Languages:** C#, Python, TypeScript, Java
- **Backend Development:** .NET, ASP.NET Core, REST APIs, microservices, dependency injection, OAuth 2.0, OpenID Connect
- **Architecture and Messaging:** Distributed systems, event-driven architecture, pub/sub, CQRS, MediatR, vertical slice architecture, Domain-Driven Design, outbox/inbox pattern, rules engines, finite state machines
- **Data and Persistence:** PostgreSQL, SQL Server, MongoDB, DynamoDB, Redis, EF Core
- **Data Platforms:** BigQuery, Google Cloud Storage, Airflow, Cloud Composer
- **Testing:** xUnit, Moq, NSubstitute, Fluent Assertions, unit testing, integration testing, test automation
- **Cloud and DevOps:** AWS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins
- **Cloud Architecture:** Cloud-native solution design, security-aware architecture, scalable integrations, reliability and operational readiness, technical discovery and design documentation
- **AWS:** AWS architecture and security, API Gateway, Cognito, IAM, ECS, ELB, Lambda, EC2, RDS, DynamoDB, S3, SNS, SQS, Bedrock, CodePipeline, CodeDeploy
- **Observability:** Datadog, Grafana, Prometheus, structured logging, monitoring, production diagnostics
- **AI Engineering:** Amazon Bedrock, SageMaker, LangChain, RAG, AI agents, MCP servers
- **Privacy and Compliance:** GDPR, CCPA, PIPEDA

Professional Experience

Solutions Architect & Senior Software Engineer — Samsung Research Canada | Vancouver, BC, Canada

Aug 2025 – Present | Contract

Projects: Samsung Credit Card Platform (USA) — Wallet Team; AI-Assisted PII Detection, Suggestion, Tagging and Data Encryption Service; AI-Based Report Generation Service

Technologies: C#, .NET, Python, Docker, PostgreSQL, Airflow, AWS, GCP, BigQuery, Google Cloud KMS, BigQuery UDFs, LangChain, RAG, MCP servers, Pinecone, Hugging Face embeddings

- Build and maintain .NET backend services, REST APIs, and wallet-platform components for a credit card product in a regulated financial-services environment.
- Participate in architecture and design discussions, translating product and technical requirements into secure, reliable services, asynchronous integrations, and data workflows using PostgreSQL, Docker, Airflow, AWS, and GCP.
- Contribute to AI-assisted workflows that identify, suggest, and tag PII using LangChain, RAG, MCP servers, Pinecone vector search, and Hugging Face embedding models to support grounded, privacy-aware handling of financial data.
- Designed and developed a Python service that detects PII in BigQuery, encrypts sensitive data with Google Cloud KMS, and applies data tags using BigQuery UDFs.
- Contribute to the development of an AI-based report generation service.
- Improve production reliability through clear service boundaries, structured logging, monitoring, operational diagnostics, and privacy-aware engineering practices.
- Communicate technical approaches, implementation trade-offs, and operational considerations through design discussions, code reviews, and technical documentation.

Senior Software Engineer — Qestrade Financial Group | Toronto, ON, Canada

Jul 2021 – Nov 2024

Projects: Stock Recommendation, QuestBank, and QuestInsurance Platforms

Technologies: C#, .NET, Python, TypeScript, Docker, PostgreSQL, MongoDB, GCP, state machines, rules engines

- Designed, developed, and maintained backend services and REST APIs supporting investment recommendations, digital banking, trading, and insurance operations.
- Built .NET and C# services, system integrations, and asynchronous event-driven components using PostgreSQL, MongoDB, Docker, and GCP.
- Implemented state machines and rules engines to automate workflows and support configurable financial-platform logic.
- Contributed to technical design discussions and cross-functional delivery, balancing business needs with security, maintainability, scalability, and production reliability.
- Applied dependency injection, repository and unit-of-work patterns, and SOLID principles to create maintainable backend systems.
- Improved reliability through production-issue analysis, database-focused development, structured logging, monitoring, and operational diagnostics.

- Contributed to technical design, code reviews, knowledge sharing, and mentoring while following financial-data privacy requirements.

Software Engineer — Konnect Technology Inc. | Penticton, BC, Canada

Oct 2018 – Jun 2021

Project: IoT-Based Factory Automation Platform for Toyota Motor Manufacturing Texas

Technologies: C#, .NET, Docker, PostgreSQL, event-driven messaging, AWS

- Built and maintained backend services for an IoT-based factory automation platform supporting industrial manufacturing operations.
- Developed .NET and C# APIs and event-driven components for device communication, automation workflows, and backend data processing.
- Used PostgreSQL, Docker, event-driven messaging, and AWS to build scalable and maintainable platform services.
- Supported production stability through troubleshooting, structured logging, backend diagnostics, code reviews, and cross-functional collaboration.

Software Engineer — Oversea Security | Seattle, WA, USA

2014 – 2018

Project: IoT-Based Marina Monitoring System

Technologies: C#, ASP.NET, Microsoft SQL Server, AWS S3, IIS

- Developed and maintained software for an IoT-based marina monitoring system.
- Built backend functionality using C# and ASP.NET, with Microsoft SQL Server for data storage.
- Integrated AWS S3 for cloud-based object storage.
- Deployed and supported the application on IIS.

Certifications

- **AWS Certified Solutions Architect – Associate**
- **AWS Certified Security – Specialty**
- [AWS Certification Verification](#)